

[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Corrective and Preventive Action (CAPA) System

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Current Version / Revision	00
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Applicable Regulatory Standards	ISO 13485:2016 Clauses 8.5.2 & 8.5.3 FDA 21 CFR Part 820.100 EU MDR 2017/745 Annex IX

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1.0 Purpose

This Standard Operating Procedure (SOP) defines the mandatory lifecycle rules for the investigation, remediation, and tracking of nonconformities, systemic process failures, adverse trends, and regulatory audit gaps. The intent of the program is to define actionable protocols to isolate the root causes of systemic quality anomalies, prevent their recurrence or occurrence, and verify the overall long-term efficacy of implementation workflows.

2.0 Scope

This procedure is universally mandatory whenever a systemic QMS failure or product anomaly occurs. It applies to critical internal and external data feeds, including recurring production nonconformities, supplier material escapes, field complaints, adverse post-market surveillance trends, internal or external regulatory audit findings, and validation study failures.

3.0 Responsibility and Authority

3.1 CAPA Owner

The engineer or department lead assigned primary operational responsibility to execute the root-cause analysis, formulate an action plan, coordinate timelines, and collect verification datasets.

3.2 CAPA Review Board (CRB)

A multi-disciplinary executive committee tasked with evaluating new CAPA requests, approving or modifying validation protocols, enforcing task deadlines, and authorizing final closure.

3.3 Quality Assurance Manager

Holds absolute authority to sign off on intermediate stages, approve risk-benefit balances for in-process changes, and verify the closure of the implementation tracking log.

4.0 Procedure

4.1 CAPA Initiation Criteria and Risk Filtering

Not all isolated nonconformities trigger a full CAPA. Initial inputs must pass through a risk-based filtering matrix using product risk classifications, frequency matrices, and clinical severity levels to prevent system overload while escalating critical safety issues.

Trigger Source	Escalation Threshold Metric	Operational Mandate
Field Complaints / Product Malfunctions	Any event triaged as a Category A or B adverse occurrence, or crossing historical baseline trend parameters.	Mandatory CAPA Log entry within 48 hours.
Regulatory Audits	Any major noncompliance finding from a Notified Body, FDA, or international inspectorate.	Mandatory CAPA Log entry within 24 hours.
Internal Manufacturing Nonconformities	Recurring batch/lot escapes for identical failure modes (e.g., >3 lots in a rolling quarter).	Formal CRB review to determine CAPA instantiation.

4.2 Phase-Gated CAPA Lifecycle Workflow

The resolution sequence follows a strict seven-stage, phase-gated lifecycle. Transition to a subsequent phase requires explicit Quality Assurance approval:

Phase 1: Identification & Containment

Clearly define the problem statement using the 5W2H methodology (Who, What, Where, When, Why, How, How Many). Immediately implement bounding containment actions to isolate suspected commercial inventory or raw stock.

Phase 2: Technical Root Cause Investigation

Utilize formal, verified engineering methodologies (e.g., 5-Whys analysis, Ishikawa/Fishbone diagrams, Fault Tree Analysis, or Design of Experiments) to isolate the true systemic failure mode.

Phase 3: Action Plan Design & Implementation

Formulate specific corrective actions (to resolve the active anomaly) and preventive actions (to eliminate the vulnerability across similar product families). Identify explicit owners, resource needs, and target timelines.

COMPLIANCE MANDATE: If an action plan requires changing product geometries, material compositions, or validated software setups, a formal change control record must be linked directly to the CAPA file.

4.3 Verification of Effectiveness (VoE) Protocol

A CAPA file cannot be closed immediately upon completing action items. The CAPA Owner must execute a predefined Verification of Effectiveness (VoE) study after a specified monitoring period (e.g., 3 to 6 months or across a minimum number of subsequent manufacturing lots). The VoE must demonstrate using objective data that the root cause has been eradicated and that no secondary risks have been introduced into production workflows.

4.4 Executive Review and File Closure

The VoE summary report is submitted to the CAPA Review Board. If the data meets the predetermined success criteria, the file status is updated to "Closed/Verified". If the failure recurs during monitoring, the CAPA must be rejected, returned to the root-cause phase, and re-investigated.

5.0 Documentation Archiving and Metrics

All CAPA files, engineering data, validation records, and CRB meeting minutes are critical quality records. They must be archived securely for a minimum of ten (10) years from file closure, and made available as primary evidence during regulatory QMS inspections.

6.0 Associated Reference Materials

- **SOP-QA-001:** Document and Data Control System
- **SOP-QA-002:** Change Control System
- **SOP-QA-003:** Complaint Handling and Post-Market Intake
- **Form-QA-006A:** CAPA Log and Investigation Worksheet

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], QA Manager	Initial baseline release of formalized Corrective and Preventive Action (CAPA) System Standard Operating Procedure.